

**School of Computer Science and Artificial
Intelligence**

WEEK-11 Assignment # 22

Program : B. Tech (CSE)
Specialization : AIML
Course Title : Competitive Programming
Course Code : 23CS302PC305
Semester : VI
Academic Session : 2025-2026
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Enrollment No. : 2303A52126
Batch No. : 33
Date : 01/04/26

Cloud Load Optimization (Power of Two Check)

Context:

A cloud server performs optimally only when the number of running processes is a power of two

Task:

Check whether the given number is a power of two using bit manipulation.

Input:

128

Output: 2

Optimal Load

Python:

```
Assignment22 > Main.py > ...
1  def is_power_of_two(n):
2      if n > 0 and (n & (n - 1)) == 0:
3          return "Optimal Load"
4      else:
5          return "Not Optimal"
6
7  # Input
8  n = int(input())
9
10 # Output
11 print(is_power_of_two(n))
```

```
PS S:\3 2 sem\CP\Week11> & C:/Users/sowmy/AppData/Local/Microsoft/Windows/Python3.11.exe "s:/3 2 sem/CP/Week11/Assignment22/Main.py"
128
Optimal Load
PS S:\3 2 sem\CP\Week11> █
```

Java:

Assignment22 > J Main.java > ...

```
1  package Assignment22;
2
3  import java.util.Scanner;
4
5  public class Main {
6      public static String isPowerOfTwo(int n) {
7          if (n > 0 && (n & (n - 1)) == 0) {
8              return "Optimal Load";
9          } else {
10             return "Not Optimal";
11         }
12     }
13
14     Run main | Debug main | Run | Debug
15     public static void main(String[] args) {
16         Scanner sc = new Scanner(System.in);
17         int n = sc.nextInt();
18
19         System.out.println(isPowerOfTwo(n));
20     }
21 }
```

```
● PS S:\3 2 sem\CP\Week11> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-XX:+S
howCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sowmy\AppData\Roaming\Code\U
ser\workspaceStorage\213ed8be1d3edcbf4523a538cc2dbf9\redhat.java\jdt_ws\Week11
_d710793f\bin' 'Assignment22.Main'
128
Optimal Load
○ PS S:\3 2 sem\CP\Week11> |
```